Are you responsible? Enhancing global explainability and reducing explanation latency in phishing email detection

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Abstract. Machine learning models for phishing email detection can achieve high accuracy. However, they often sacrifice explainability for predictive performance and are therefore treated as black boxes. Existing explainable AI (XAI) techniques typically focus on explaining individual predictions, but they often lack global interpretability and can be computationally expensive. This study aims to improve the global interpretability of phishing detection models while reducing the cost of generating local explanations. To achieve this, we use global surrogate models to approximate and explain the behavior of the original models. In particular, we adopt Explainable Boosting Machine (EBM) and GAMI-Net as global surrogate models for Random Forest and Deep Neural Network teacher models. We evaluate the surrogate models using accuracy fidelity, F1 fidelity, and a proposed Error Fidelity metric, which measures the ability of the surrogate to explain the original model's unexpected behavior. Experimental results show that both EBM and GAMI-Net achieve high fidelity for both teacher models.

1 Introduction

As the digital world is evolving, cyber-crime especially email phishing is a threat to the internet users. Current techniques to detect phishing are primarily based on machine learning models that are reliable in terms of accuracy, however, it's trustworthy is not definite. Sarker et al. (2024) stated that the transparency of the model contributes to its worthiness and accountability. Phillips et al. (2020) also reinforce the importance as an intelligence needs to be explainable, meaningful, reflect on its output and acknowledge it's limitation. Hence a global explanation of a model can gain trust from businesses, customers, help diagnose incidents, and acknowledge its weak points. This assists the models to aligns with the principles identified in Phillips et al. (2020).

However, current techniques of explainable AI study on email phishing detection models are only interpret for single email. This helps the model responsible for its own output, but it does not directly help the model meaningful overall. Moreover,the latency can be a pain point since current techniques requires expensive operations.

Therefore, our we aims to uncover the black-box nature of the models then give a global explanation and reduce the latency to compute explanation for individual email.

2 Related Work

2.1 XAI methods to explain email phishing detection models

Traditional machine learning algorithms has been widely used to detect phishing emails. Safi and Singh (2023) analysed that different machine learning algorithms dominants in email phishing detection methods. Algorithms such as Random Forest, Support Vector Machine or Multi-layer Perceptron are identified to be studied in vast number of studies. However, Radulovic et al. (2021) argues that machine learning model such as Random Forest and Neural Networks are a black-box classifiers. Current techniques for explaining phishing email detection model shown in Sharma et al. (2025) shown that LIME and SHAP are two most common methods to explain a black box model in cybersecurity, specifically email phishing. However, these methods are mainly classified as local post-hoc methods which only produce interpretation per sample instead of explain an overall behavior.

2.2 Global Post-hoc Model Agnostics Methods

There are several global post-hoc model agnostics methods that we can adopt to explain email phishing models. Friedman and Popescu (2008) used Rule Ensembles to build an intrinsically explainable models using if-then rules. However, if the complexity grows, explainability can be reduced and it is instable. Apley and Zhu (2020) on the other hand used Accumulated Local Effects which accumulate small changes to observe changes in the predictions. However, the authors also state that it may be instable and does not scale well. Bastani et al. (2018) used a decision tree as a global surrogate models to interpret the original models. However, the author also, pointed out the simple models such as decision tree may not fit well on the training data with low-depth tree.

2.3 Glass-box models

To address the under-fitting problem of simple global surrogate models. Alternative choices for global surrogate models can be adopted. Generalized Additive Models (GAM) is a family of additive models that can be interpretable and visualized (Wood, 2017).

3 Problem definition

3.1 Problem formulation

We study global post-hoc, model-agnostic explanation for phishing email detection. Let the original labelled dataset be $D = \{(x_i, y_i)\}_{i=1}^{N}$, where $x_i \in R^d$

denotes the processed feature vector of an email and $y_i \in \{0, 1\}$ denotes the ground-truth class label, with $y_i = 1$ indicating phishing and $y_i = 0$ indicating legitimate email. Let P_X denote the input distribution over the feature space $\mathcal{X} \subseteq R^d$.

A black-box teacher model F is first trained for phishing detection. Rather than directly explaining its internal parameters, our aim is to learn an interpretable global surrogate model S whose behaviour approximates that of the teacher over the input distribution. Formally, the goal is to obtain a surrogate such that

$$S(x) \approx F(x), \quad \forall x \sim P_X.$$

3.2 Dataset collection

The dataset was constructed by integrating multiple publicly available phishing and legitimate email corpora into a unified business-email dataset for phishing detection model training. These sources include Nazario Phishing Corpus (Nazario, 2007), Kaggle phishing datasets (Kaggle, 2024), and the Enron Email Dataset (Cohen, 2015). In total, 132,289 samples are collected.

3.3 Dataset analysis and preparation

The dataset was analysed to characterise its class composition, attack-type coverage, missingness, and the distribution of key behavioural indicators relevant to phishing detection. Following dataset construction, records were cleaned, deduplicated, and standardised into a unified schema containing message content, sender metadata, URLs, attachments, class labels, and derived indicators such as link presence, urgent language, reply-to mismatch, and suspicious sender patterns. This analytical stage served two purposes: first, to validate the quality and consistency of the integrated business-email corpus; and second, to identify which observable properties most plausibly differentiate phishing from legitimate emails.

Descriptive analysis focused on label distribution, attack-type distribution, missing values, average body length, mean number of links, prevalence of urgent wording, and body-length differences between phishing and legitimate classes. These summaries were used to assess potential class imbalance, verify that the dataset captured multiple business-phishing behaviours, and confirm that the engineered variables aligned with the intended threat patterns. In the modelling pipeline, additional engineered lexical, structural, sender-domain, URL, and attachment features were extracted before splitting the data into training, validation, and test subsets in a 70:10:20 ratio. Numerical variables were imputed and scaled, while categorical variables were encoded according to cardinality, producing a processed feature space of 292 variables for downstream modelling and interpretability analysis.

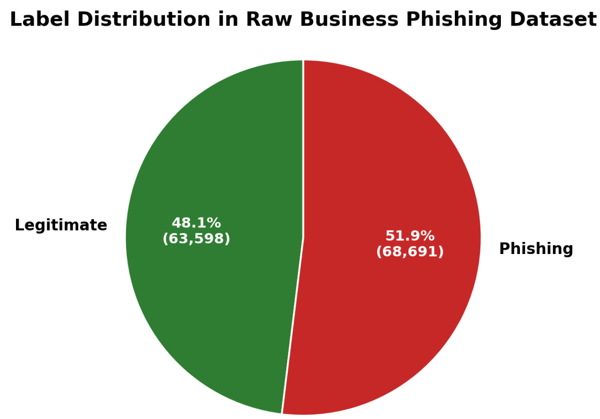

Fig. 1: Label distribution

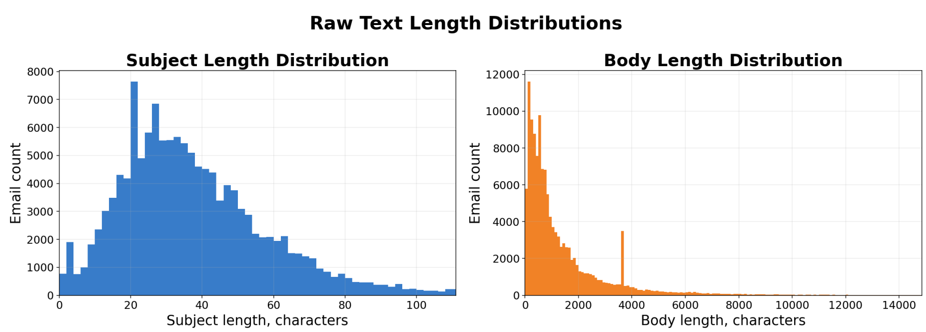

Fig. 2: Subject and body length distribution

attribute	mean	median	std	min	max
label	0.52	1.0	0.50	0.0	1.0
num_links	1.65	0.0	10.29	0.0	3133.0
has_links	0.47	0.0	0.50	0.0	1.0
num_attachments	0.00	0.0	0.00	0.0	0.0
has_attachment	0.00	0.0	0.00	0.0	0.0
has_urgent_words	0.09	0.0	0.29	0.0	1.0
sender_replyto_mismatch	0.00	0.0	0.00	0.0	1.0
suspicious_sender_domain	0.04	0.0	0.19	0.0	1.0
suspicious_attachment_type	0.00	0.0	0.00	0.0	0.0
subject_length	38.12	34.0	31.90	0.0	7169.0
body_length	1857.21	790.0	12460.84	1.0	1829070.0

Table 1: Summary statistics of dataset attributes

4 Method

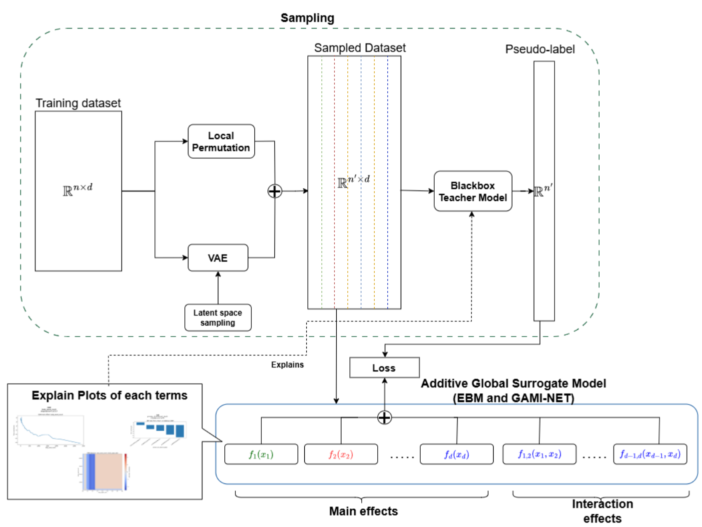

Fig. 3: Overview of the methodology

4.1 Global additive surrogate models

To approximate the behavior of the black box model, we adopt a global surrogate model that is intrinsically intepretable and behaves approximately similar to the original models. We define original black-box model as $F(x)$ and global surrogate model as $S(x)$. If two models behave similarly, their prediction should align with each other. Therefore, our learning objective is:

$$S^* = \arg\min_{S \in \mathcal{S}} \ \mathcal{L}(S; D')$$

Where D' is the training dataset for $S(x)$ defined as $D' = \{(x_i, F(x_i))\}_{i=1}^{|D'|}$. Hence, the goal is to minimize the loss function:

$$\mathcal{L}(S; D') = \frac{1}{|D'|} \sum_{x_i \in D'} -\Big[F(x_i) \log S(x_i) + \big(1 - F(x_i)\big) \log \big(1 - S(x_i)\big)\Big]$$

For the choice of Global Surrogate Model, we adopt a family of generalized additive models (Hastie & Tibshirani, 1990). These models include Explainable Boosting Machine (Lou et al., 2013) and GAMI-NET (Yang et al., 2021). A generalized additive model is an additive model where each term only predicts the output based on one feature only for main effects. Therefore, the contribution for each feature can be plotted on a 2D plot. Another optional term that can be added is the interaction terms where terms predict using 2 features. Hence, it can be a 3D plots or a heatmap. The definition for the Global additive surrogate model is:

$$logit(S(x)) = \beta_0 + \sum_k f_k(x_k) + \sum_{i<j} f_{ij}(x_i, x_j)$$

Where $f_k(x_k)$ is a term that learn on 1 feature only and $f_{i,j}(x_i, x_j)$ is a term learn on the interaction terms of 2 features. For EBM, each term is a decision tree while for GAMI-Net is a sub-neural-network.

4.2 Sampling Method

Due to the curse of dimensionality, the space for sampling grow exponentially with number of dimensionalities. However, the necessary data samples for training surrogate model are lying near the decision boundary of the original model. Because original model are trained on the original training dataset, model's decision structure is influenced by the training data distribution, so sampling near the data distribution may represent the decision behaviour of the original model. Hence, we adopt 2 method of sampling includes local perturbations and Variational Auto Encoder (VAE) sampling.

Local perturbation This method of sampling will randomly select data samples from training dataset and apply gaussian noise scaled by the standard deviation of a cloud of k nearest neighbor samples to generate new synthetic data samples.

$$\tilde{x} = x + \lambda\epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma_k^2)$$

VAE sampling This method of sampling will train a VAE on the original dataset to learn the underlying data distribution in the latent space. New synthetic data will be generated by sampling in the latent space of VAE. This enable broader coverage of data distribution beyond local perturbations.

$$z \sim \mathcal{N}(0, I), \qquad \tilde{x} \sim p_\theta(x \mid z)$$

5 Experimental Setup

5.1 Data Preparation

First the emails text field are extracted meaningful features by feature engineering into fields such as "Urgent word counts", "Length". Then missing values are imputed which numerical data is imputed by median and normalize with standardization while numerical data is imputed with most frequent value. For high-cardinality categories, they are label encoded while low cardinality categories are one-hot encoded. After the data processing, the dataset results in 292 features. With high dimension, it contributes to the low interpretability of the models.

5.2 Blackbox teacher setup

In this experiment, two teacher models are considered. The first model is Random Forest (RF) and the second model is Deep Neural Network (DNN). Random Forest teacher model is implemented with 300 trees with random state 42 for reproducibility. Because Random Forest implemented in this experiment use 300 trees and the prediction output is a collective decision of all trees, global and local understanding is hard to achieve. Deep Neural Network in this experiment uses 2 hidden layers 128 and 64 neurons with ReLu activation function. This implies the model has 45825 learnable parameters. Although the model is extremely deep, with the non-linearity and complex feature interactions, the model behaves as a black-box hence global and local understanding is hard to achieve.

5.3 Sampling Setup

Local perturbation sampling applied to real training samples for the teacher models. For each sample, the k-nearest-neighbor cloud was set at k = 25. Gaussian noise is added with using the standard deviation with scale 0.15. Then, for one-hot and label encoded features, the value is floor down and ground to nearest whole value within the range of training samples. As the result, 300 000 instances are sampled. Variational Auto Encoder use in this experiment configured with latent space of 20 dimensions and hidden dimensions of 256, KL-divergence weight $\beta = 1.0$, learning rate of 0.001. For sampling, the temperature is set at 0.85. Similar to local perturbation method. Processed categorical data is floor and grounded accordingly.

5.4 Global Surrogate model setup

EBM setup in this experiment uses interactions = 5, validation size=0.15, outer bags=8, inner bags=0, learning rate=0.03, max rounds=5000, early stopping rounds=100, max bins=256, max interaction bins=64, n jobs=-1, and random state=42. This setup allows EBM to achieve higher fidelity with interaction term whiles maintaining interpretability.

GAMI-Net is configured as interact num=5, subnet arch=[40,40,40,40,40], interact arch=[40,40,40,40,40], batch size=512, main effect epochs=50, interaction epochs=50, tuning epochs=20, backpropagation learning rate =[0.0001 , 0.0001 , 0.0001], early stop threshold=[30,30,20], heredity=True, reg clarity=0.1, validation ratio=0.15, and random state=42. For interpretation, top 10 important features of both EBM and GAMI-NET is plotted with top 5 interaction term.

5.5 Evaluation Metrics

Primary Evaluation Metrics

Accuracy Fidelity Accuracy Fidelity measures how accurately the prediction produce by the surrogate model on original model predictions (Andrews et al., 1995). A higher accuracy fidelity indicates stronger behaviours agreement between surrogate and original model.

$$\text{Fidelity}_{\text{Acc}}(S, F) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[S(x_i) = F(x_i)\right]$$

where S is the surrogate model and F is the original model.

F1 Fidelity Similar to Accuracy Fidelity also measure how accurate if the alignment of the prediction of surrogate model and original mode while accounting for both precision and recall across the predicted classes (Andrews et al., 1995).

$$\text{Fidelity}_{\text{F1}}(S, F) = F_1\left(\{F(x_i)\}_{i=1}^{N}, \{S(x_i)\}_{i=1}^{N}\right)$$

Proposed supplemental metric – Error Fidelity Although accuracy and F1 fidelity is strong in measuring how faithful the surrogate model compares with the original model, it may be weak in measuring the capability of explaining teacher unexpected behaviour. Consider the following example:

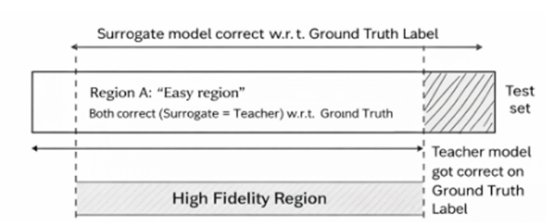

Fig. 4: Surrogate model achieves high fidelity but does not agree with the teacher model on teacher unexpected behavior.

Although the surrogate achieved high fidelity with original model, it failed to agree with original model on teacher errors on the ground truth label. According to Phillips et al. (2020), to gain social acceptance, if the output is unexpected from the system, the explanation may need to assist users to understand the reason. Therefore, this experiment proposes Error fidelity which measures the accuracy fidelity given that the output of original model is mismatched with ground truth.

Let the error set be defined as:

$$\mathcal{E} = \{x_i \mid F(x_i) \neq y_i\}$$

Then the error fidelity is defined as:

$$\mathrm{ErrFid}(S, F) = \frac{1}{|\mathcal{E}|} \sum_{x_i \in \mathcal{E}} \mathbf{1}\big[S(x_i) = F(x_i)\big]$$

6 Result

6.1 Teacher performance on test set

Teacher	Accuracy	Precision	Recall	F1
Random Forest	**0.9433**	**0.9524**	**0.9377**	**0.9450**
Deep Neural Net	0.9149	0.9168	0.9195	0.9182

Table 2: Performance of teacher models on the test set.

6.2 Accuracy and F1 fidelity of each surrogate to its teacher

Teacher	Surrogate	Accuracy	F1
Random Forest	EBM	**0.9260**	**0.9280**
Random Forest	GAMI-Net	0.8934	0.8936
Random Forest	Logistic (Base Line)	0.8013	0.8093
Deep Neural Net	EBM	**0.9402**	**0.9425**
Deep Neural Net	GAMI-Net	0.9137	0.9142
Deep Neural Net	Logistic (Base Line)	0.8093	0.8093

Table 3: Accuracy and F1 fidelity of each surrogate model with respect to its teacher model.

From table 2, it can be obeserved that EBM and GAMI-Net achieved high fidelity with both Random forest and Deep Neural Net. Comparing with baseline of

Logistic regression, both models shows significant dominance, suggesting that both models picked up more complex behaviour of original models than just a linear relationship. Moreover, EBM outweights GAMI-Net on both teachers, suggesting EBM are more capable of learning the teacher behaviour.

6.3 Supplemental metric - Error Fidelity

Teacher	Surrogate	Teacher Error Count	Teacher Error Rate	Error Fidelity
Random Forest	EBM	1500	0.0567	**0.7673**
Random Forest	GAMI-Net	1500	0.0567	0.7367
Random Forest	Logistic (Baseline)	1500	0.0567	<u>0.6373</u>
Deep Neural Net	EBM	2252	0.0851	**0.7695**
Deep Neural Net	GAMI-Net	2252	0.0851	0.7575
Deep Neural Net	Logistic (Baseline)	2252	0.0851	<u>0.6803</u>

Table 4: Error Fidelity comparison across surrogate models.

Although both EBM and GAMI-Net achieved high fidelity from table 2. Their error fidelity is at acceptance rate which about is about 75%. More specifically, EBM still outweights GAMI-Net by a slight margin with 0.7673 to 0.7367 and 0.7695 to 0.7575. Both models also dominant the baseline of logistic regression.

6.4 Ablation Study Result

Teacher	Dataset	EBM Error Fidelity	GAMI-Net Error Fidelity
Random Forest	92k raw	0.7507	0.7173
Random Forest	300k perturbation	<u>0.7180</u>	<u>0.7073</u>
Random Forest	300k+VAE	**0.7673**	**0.7367**
Deep Neural Net	92k raw	0.7496	0.7651
Deep Neural Net	300k perturbation	<u>0.7194</u>	**0.7682**
Deep Neural Net	300k+VAE	**0.7695**	<u>0.7575</u>

Table 5: Ablation study on different data generation strategies.

We did ablation study on different data samples strategies where we remove part of our data samples use for training global surrogate models. From table 4, it can be observed that if only local perturbation is used. the error fidelity may be hurt but mixing perturbation and VAE samplings, it can help improve the error fidelity.

6.5 Interpretation Plots

EBM vs GAMI-Net on Deep Neural Network Teacher

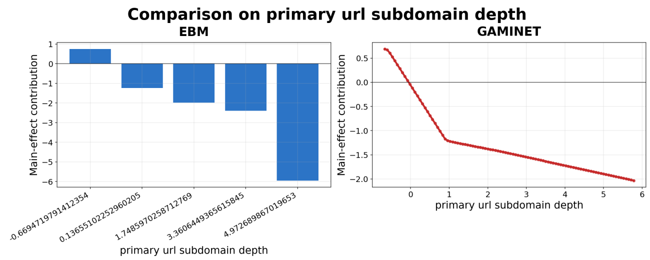

Fig. 5: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on primary subdomain depth

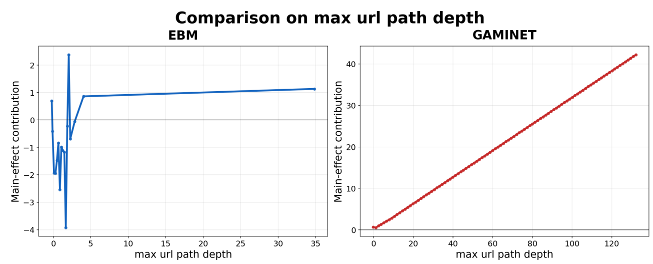

Fig. 6: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on max url path depth

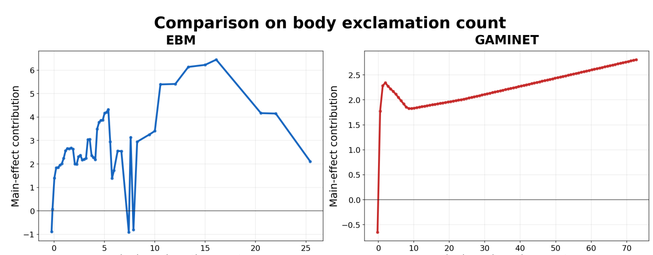

Fig. 7: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on exclamation count

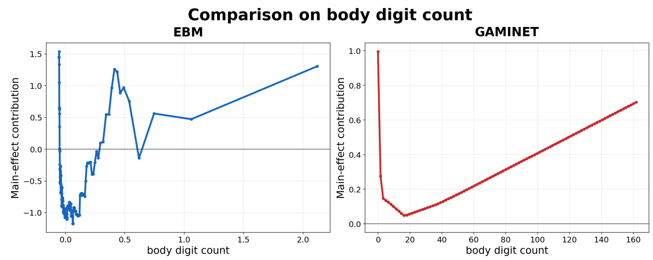

Fig. 8: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on body digit count

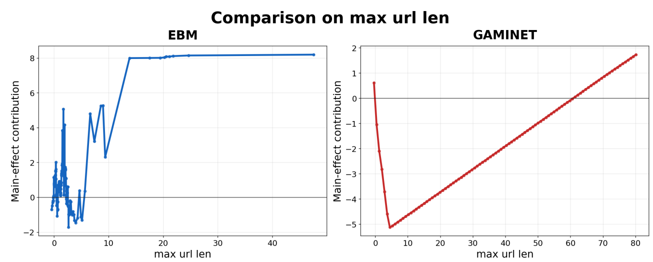

Fig. 9: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on max url length

EBM vs GAMI-NET on Random Forest Teacher

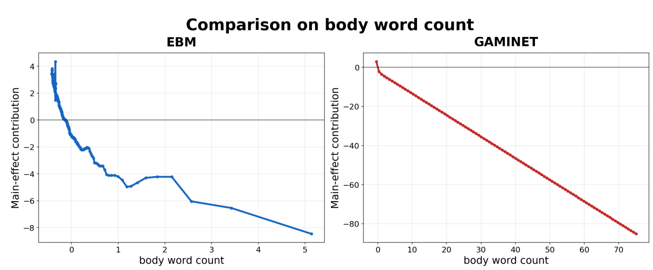

Fig. 10: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on body word count

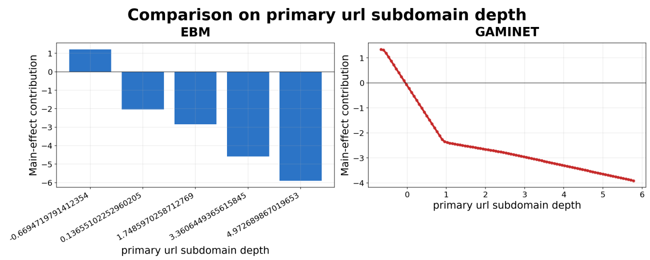

Fig. 11: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on url subdomain depth

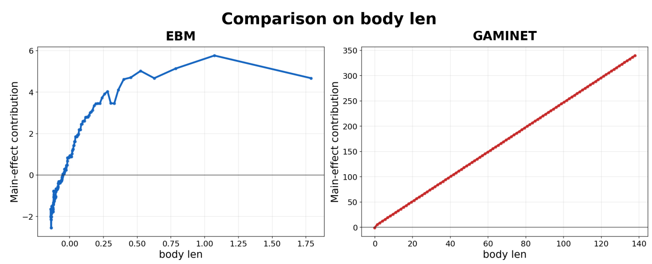

Fig. 12: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on body length

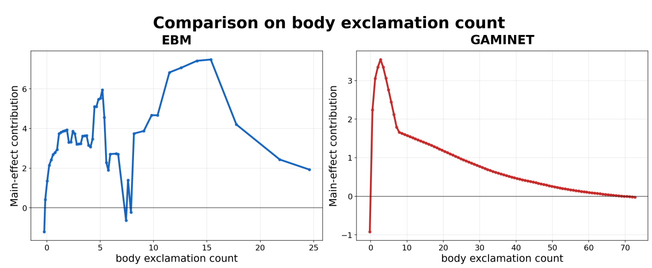

Fig. 13: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on exclamation count

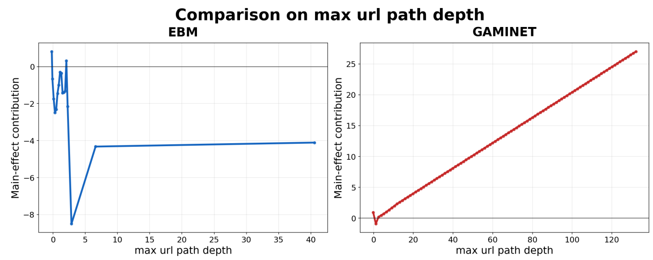

Fig. 14: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on max url path depth

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